Quadratic Equations

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 80 minutes · Total marks: 80

KEY RULES

Factorising	Rearrange to $= 0$, factorise, set each bracket $= 0$
Formula	$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Completing the square	$x^2 + bx = \left(x + \dfrac{b}{2}\right)^2 - \left(\dfrac{b}{2}\right)^2$
Discriminant	$b^2 - 4ac > 0$: two roots; $= 0$: one; < 0: none

WORKED EXAMPLE 1

[3 marks]

Solve $x^2 - x - 12 = 0$

Step 1	Factorise	$(x - 4)(x + 3) = 0$
Step 2	Set each bracket to zero	$x - 4 = 0$ or $x + 3 = 0$

Answer: $x = 4$ **or** $x = -3$

WORKED EXAMPLE 2

[3 marks]

Solve $2x^2 + 5x - 3 = 0$

Step 1	$a \times c = -6$: numbers 6 and -1	$2x^2 + 6x - x - 3 = 0$
Step 2	Factorise in pairs	$2x(x + 3) - 1(x + 3) = 0$
Step 3	Write the brackets	$(2x - 1)(x + 3) = 0$

Answer: $x = \dfrac{1}{2}$ **or** $x = -3$

WORKED EXAMPLE 3

[4 marks]

Solve $x^2 + 4x - 3 = 0$**. Give your answers to 2 decimal places.**

Step 1	Use the formula with $a = 1,\ b = 4,\ c = -3$	$x = \dfrac{-4 \pm \sqrt{16 + 12}}{2}$
Step 2	Evaluate the square root	$x = \dfrac{-4 \pm 5.2915}{2}$
Step 3	Work out both values	0.6458 or -4.6458

Answer: $x = 0.65$ **or** $x = -4.65$

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1 Solve $x^2 - 5x = 0$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 Solve $25t^2 - 16 = 0$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 Solve $2x^2 + 7x + 3 = 0$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

4 Solve $x^2 = 5x + 6$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

5 Solve $x + \frac{6}{x} = 5$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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6 Solve $3x^2 - x - 2 = 0$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

7 (i) Factorise $x^2 + 2x - 48$

.....

(2)

i) Hence, solve $x^2 + 2x - 48 = 0$

.....

(1)

(Total for Question 7 is 3 marks)

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8 (i) Factorise $x^2 + 2x - 24$

.....

(2)

ii) Hence solve $x^2 + 2x - 24 = 0$

.....

(1)

(Total for Question 9 is 3 marks)

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9 Solve $x^2 - 21x + 20 = 0$
Show your working clearly.

.....

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10 (i) Factorise $x^2 + 9x - 22$

.....

(2)

(ii) Hence, solve $x^2 + 9x - 22 = 0$

.....

(1)

(Total for Question 11 is 3 marks)

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11 (i) Factorise $x^2 + 5x - 24$

.....

(2)

i) Hence, solve $x^2 + 5x - 24 = 0$

.....

(1)

(Total for Question 11 is 3 marks)

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12 Solve $3x^2 + 2x - 7 = 0$
Give your solutions correct to 3 significant figures.
Show your working clearly.

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13 Solve the equation $5x^2 + 8x - 23 = 0$
Show your working clearly.
Give your solutions correct to 3 significant figures.

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14 Solve $5x^2 + 2x - 4 = 0$
Give your solutions correct to 3 significant figures.
Show your working clearly.

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15 **20** Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \dfrac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c
Show your working clearly.

$a =$ .....

$c =$ .....

(Total for Question 20 is 3 marks)

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16 Here is a trapezium.

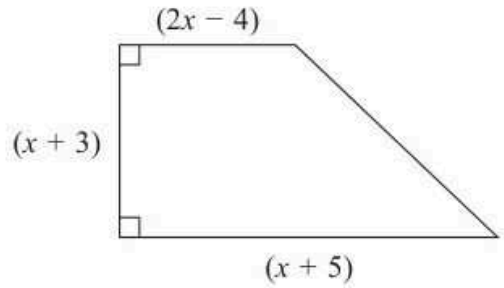


Diagram **NOT** accurately drawn

All measurements are in centimetres.

The area of the trapezium is $60\,cm^2$

(a) Show that $3x^2 + 10x - 117 = 0$

(3)

(b) Work out the value of x
Show your working clearly.
Give your answer correct to 3 significant figures.

(3)

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17 17

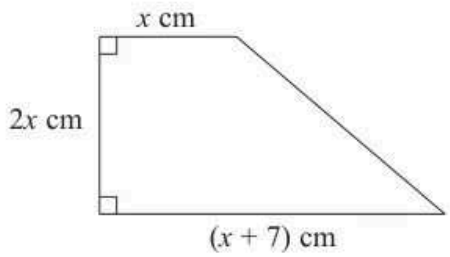


Diagram **NOT** accurately drawn

The diagram shows a trapezium.
The trapezium has an area of 17 cm^2

(a) Show that $2x^2 + 7x - 17 = 0$

(3)

(b) Work out the value of x.
Give your answer correct to 3 significant figures.
Show your working clearly.

$x =$ .....

(3)

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CHALLENGE QUESTIONS

18 The diagram shows a trapezium.

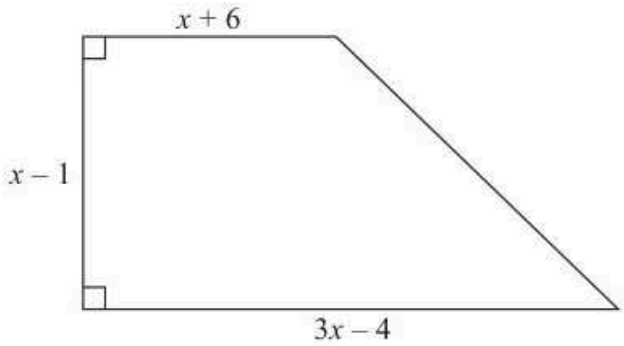


Diagram **NOT** accurately drawn

All measurements on the diagram are in centimetres.

The area of the trapezium is 119 cm^2

(i) Show that $2x^2 - x - 120 = 0$

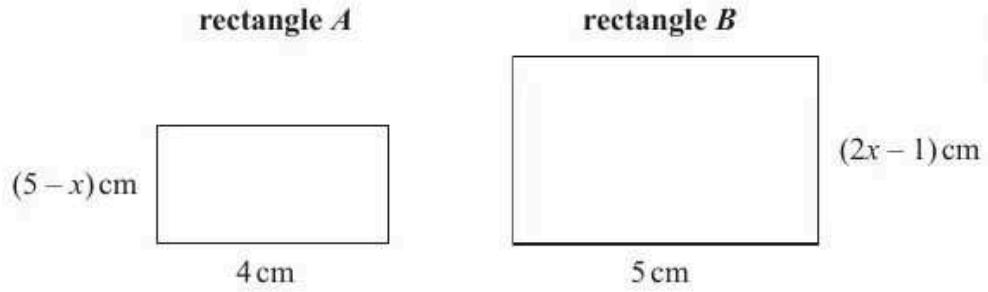
(ii) Find the value of x.
Show your working clearly.

$x =$ .....

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19 Here are two rectangles, rectangle A and rectangle B.



The area of rectangle B is twice the area of rectangle A.

Work out the value of x.
Show your working clearly.

$x =$ .....

(Total for Question 2 is 4 marks)

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20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

.....

(Total for Question 20 is 4 marks)

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21 23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

$k =$ .....

(Total for Question 23 is 5 marks)

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22 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are (0.2, 2.6)

Work out the coordinates of B
Show your working clearly.

(..... , ...)

(Total for question = 6 marks)

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TOTAL FOR PAPER IS 80 MARKS